APEX ERP Enterprise

Complete Step-by-Step Operations & Workflow Guide

Multi-Location Inventory, Warehouse Topology, Double-Entry Stock Ledger, Procurement Approvals & RBAC Security Governance

SYSTEM RELEASE
v1.0.0 Enterprise Warehouse Edition

FRONTEND WEB PORTAL
React 18 + Vite (Port 5173)

BACKEND API SERVICE
Express + TypeScript + Prisma ORM (Port 5001)

ACID DATABASE
PostgreSQL 15 (Port 5434)

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1.0 System Architecture & 1-Click Login Personas

The **Apex Enterprise ERP** is engineered for high-precision multi-location supply chain management. It enforces strict ACID compliance where every physical inventory movement writes an immutable debit/credit transaction to the stock ledger.

> **Default Credentials:** All seeded accounts use the universal password `Password123!`. You can switch between demo personas instantly from the top navigation bar or the login screen.

PERSONA	USERNAME	SYSTEM ROLE	PRIMARY FUNCTIONAL RESPONSIBILITIES
System Administrator	`admin`	`ADMIN`	Full global rights, user management, role assignments, system-wide overrides.

PERSONA	USERNAME	SYSTEM ROLE	PRIMARY FUNCTIONAL RESPONSIBILITIES
Inventory Manager	`inv_manager`	`INVENTORY_MANAGER`	Product catalog master, UoM conversions, reorder thresholds, variance reconciliation.
Warehouse Manager	`wh_manager`	`WAREHOUSE_MANAGER`	Warehouse topology, zones, racks, storage bins, inter-facility transfer dispatch.
Warehouse Staff	`wh_staff`	`WAREHOUSE_STAFF`	Dock receiving (GRN), physical bin putaway, transfer intake, barcode counts.
Procurement Officer	`proc_officer`	`PROCUREMENT_OFFICER`	Supplier master records, purchase order generation, contractual terms.
Department Head	`dept_manager`	`DEPARTMENT_MANAGER`	Initiates and submits departmental purchase requisitions (PR) for operational needs.
Finance Approver	`fin_approver`	`FINANCE_APPROVER`	Budget line verification, capital expenditure authorizations, final sign-off.
Internal Auditor	`auditor`	`AUDITOR`	Strict read-only compliance inspection, immutable ledger review, before/after JSON diffs.

2.0 Role-Based Access Control (RBAC) Governance

The ERP features a granular permission engine. Each role is bound to a defined subset of the 25+ permissions seeded in the PostgreSQL database. Separation of duties prevents conflict of interest (e.g., users cannot approve their own purchase requisitions).

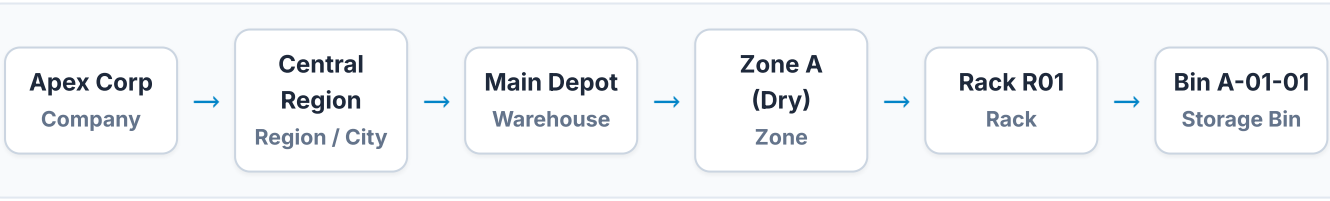
How to Manage Users & Review Permission Matrix (/users)

`ADMIN ONLY`

1. Log in as `admin` and navigate to **Users & Permissions** from the sidebar.
2. **User Directory Tab:** Search by employee name, corporate email, or department. Review assigned warehouse scopes and status.
3. **Suspend / Reactivate:** Click the `Suspend` button on any active user to revoke immediate API token validation.
4. **Role Assignment:** Click `+ Role` to add secondary roles or remove existing role badges.
5. **Permission Matrix Tab:** Review the interactive matrix showing exactly which granular gates (e.g., `inventory:write`, `procurement:approve`) each role possesses.

3.0 Workflow 1: Warehouse & Bin Hierarchy Setup

Storage locations are modeled in an 8-level physical topology: **Company** → **Region** → **City** → **Site** → **Warehouse** → **Zone** → **Rack** → **Storage Bin**.



Step-by-Step Topology Setup Instructions

`WAREHOUSE_MANAGER`

1. Navigate to **Warehouse Tree** (`/hierarchy`) in the sidebar.
2. Expand any node to inspect child facilities, warehouses, and storage zones.
3. Click on any **Bin Code** (e.g., `BIN-A-01-01`) in the tree. The right-hand inspector will instantly display:
   - Bin storage type (`STORAGE`, `STAGING`, or `COLD_STORAGE`).
   - Physical on-hand items, batches, and unit quantities currently occupying the bin.
4. To expand warehouse capacity, click **Add Bin**, select the target facility, zone, and rack, then specify the unique bin barcode.

4.0 Workflow 2: Product Master Catalog Management

Standardized item records ensure accurate replenishment thresholds, barcoding, and automated low-stock warnings.

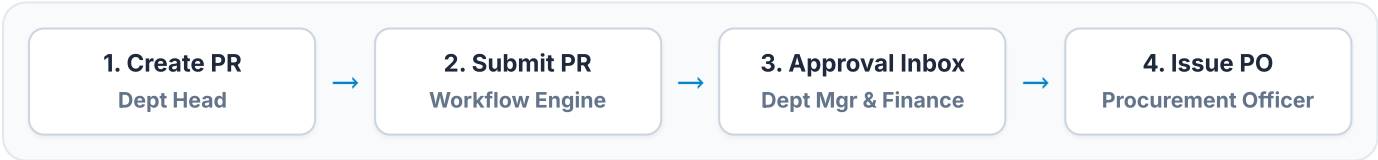
Adding and Managing SKUs (/products)

`INVENTORY_MANAGER`

1. Navigate to **Product Master** (`/products`).
2. Click **+ New Product** to open the item creation dialog.
3. Fill in required parameters:
   - **SKU Code:** Unique product identifier (e.g., `SKU-BRG-200`).
   - **Product Name:** Descriptive item label (e.g., Precision Ball Bearing 12mm).
   - **Category & UoM:** Select item group and unit (Pieces, Kilograms, Liters).
   - **Inventory Thresholds:** Set `Min Stock` (10), `Max Stock` (500), and `Reorder Level` (25).
   - **Financial Values:** Enter standard Cost Price ($) and Selling Price ($).
4. Save the record. The system automatically computes stock status badges: `NORMAL` , `LOW STOCK` , or `OUT OF STOCK` .

5.0 Workflow 3: Procurement & Approval Lifecycle

Procurement follows a rigorous three-step gate: **Requisition → Multi-Tier Workflow Approval → Purchase Order**.



Phase 1: Submitting Purchase Requisitions (/procurement)

`DEPARTMENT_MANAGER`

1. Log in as `dept_manager` and visit **Procurement & PO**.
2. Click **New Requisition**. Select Department, Product, Requested Quantity, and provide justification.
3. On the draft row, click **Submit Approval**. This transitions the document into `SUBMITTED` status and invokes the sequential approval engine.

Phase 2: Reviewing in Approval Inbox (/approvals)

`FINANCE_APPROVER`

1. Switch persona to `fin_approver` . Notice the inbox counter badge updates.

2. Visit **Approval Inbox** (`/approvals`).
3. Inspect the requisition details, justification, and estimated total expenditure.
4. Choose one of three authorized actions:
    - **Approve Step:** Advances workflow to final approval and updates underlying PR to `APPROVED` .
    - **Return for Edit:** Sends document back to draft for the initiator to revise quantities.
    - **Reject:** Immediately closes the requisition with mandatory reason comments.

Phase 3: Generating Vendor Purchase Order (/procurement) `PROCUREMENT_OFFICER`

1. Switch persona to `proc_officer` .
2. Click **Create Purchase Order**. Choose registered Supplier, Product SKU, contracted unit price, and payment terms.
3. Save the PO. It is assigned a tracking number (e.g., `PO-2026-001`) and set to `ORDERED` .

6.0 Workflow 4: Goods Receiving Dock & Bin Putaway

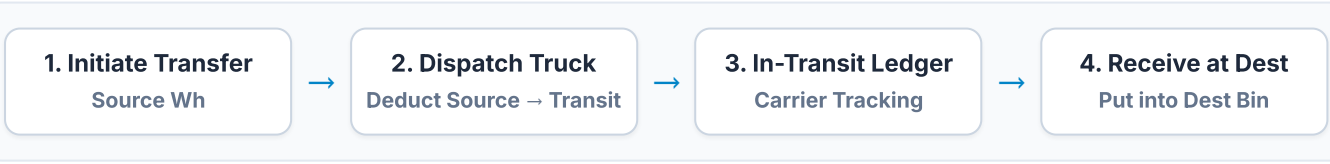
Receiving at the dock does not immediately increase general sellable balance until items pass quality check and are placed in specific bins.

Dock Receiving & Bin Putaway Procedure (/receiving) `WAREHOUSE_STAFF`

1. Log in as `wh_staff` and open **Receiving & Putaway**.
2. **Generate GRN (Goods Receipt Note):** Click `Generate GRN` when the delivery truck arrives.
   - Select Supplier, Receiving Warehouse facility, Product, and delivered quantity.
   - Enter physical inspection breakdown: `Accepted Qty` (e.g., 95) vs `Damaged Qty` (e.g., 5).
   - Specify the supplier's Batch/Lot number for traceability.
3. **Automated Putaway Task:** The system automatically writes the GRN and generates a pending putaway task for the accepted units, targeting the optimal bin.
4. **Confirm Bin Placement:** In the *Pending Putaway Tasks* tab, review the target bin code (e.g., `BIN-A-01-02`) and click **Confirm Putaway to Bin**.
5. **ACID Ledger Posting:** The system atomics writes a `RECEIPT` posting to the double-entry stock ledger, crediting available balance in that specific bin.

7.0 Workflow 5: Inter-Warehouse Stock Transfers

Moving stock between distant hubs utilizes an in-transit buffer ledger to ensure zero loss or double-counting during transit.



Executing Inter-Facility Transfers (/transfers) `WAREHOUSE_MANAGER & STAFF`

1. Visit **Stock Transfers** (`/transfers`).
2. Click **Initiate Transfer**. Select Source Warehouse, Destination Warehouse, Product, and requested units.
3. **Dispatching:** When freight departs, click **Dispatch Stock**. Enter carrier name and freight tracking code. Source on-hand balance is immediately debited, and units move to `IN_TRANSIT`.
4. **Destination Intake:** When the truck arrives at the receiving facility, the destination staff clicks **Receive Goods**, selects the local destination bin, and records actual count.

5. **Discrepancy Ledger:** If received units are less than dispatched, the system flags a discrepancy shortage and triggers an audit log.

8.0 Workflow 6: Physical Cycle Counting & Reconciliation

Cycle counting reconciles system balances against physical reality without stopping warehouse operations.

Cycle Count & Variance Reconciliation (/stock-counts) `INVENTORY_MANAGER & STAFF`

1. Visit **Counts & Variance** (`/stock-counts`).
2. Click **New Cycle Count**. Select Target Warehouse and methodology (Cycle Count, Full Inventory, Spot Check).
3. The system captures an immutable snapshot of current balances across all bins.
4. **Record Physical Counts:** Staff walks aisles and records counted physical items. Select reason code if variance is detected (`SURPLUS` , `LOSS` , `DAMAGE`).
5. **Post Adjustments:** The Inventory Manager reviews computed variances and clicks **Post Adjustments**. The system posts atomic `ADJUSTMENT_UP` or `ADJUSTMENT_DOWN` ledger records, synchronizing balances.

9.0 Workflow 7: Valuation Reports & Audit Trail

Valuation Intelligence & CSV Export (/reports) `FINANCE & AUDIT`

1. Navigate to **Reports & Analytics** (`/reports`).
2. Review the total inventory valuation card, calculated dynamically using standard cost across all warehouses.
3. Click **Export CSV** to download a formatted spreadsheet report for accounting records.

Immutable Compliance Audit Trail (/audit) `AUDITOR`

1. Navigate to **Audit Trail** (`/audit`).
2. Filter by entity (`StockMovement` , `PurchaseOrder` , `User`).
3. Click **View Diff** on any row to open the modal showing side-by-side JSON snapshots of the system state before and after the modification.

10.0 Theme Customization & Database Inspection

ACTION	PROCEDURE & INSTRUCTIONS
Theme Switcher	Click the **Sun / Moon** toggle button in the top navbar or login page. Toggles seamlessly between Light Mode (crisp slate contrast) and Dark Mode (executive dark glassmorphism). Preference is saved in browser storage.
Prisma Studio (Web GUI)	Open terminal in backend and run: `node node_modules/prisma/build/index.js studio` Open browser at **http://localhost:5555** to visually explore all 56 PostgreSQL tables.
External DB Tool	Connect DBeaver / pgAdmin to `localhost:5434` with user `erp_user`, password `erp_password`, database `enterprise_erp`.